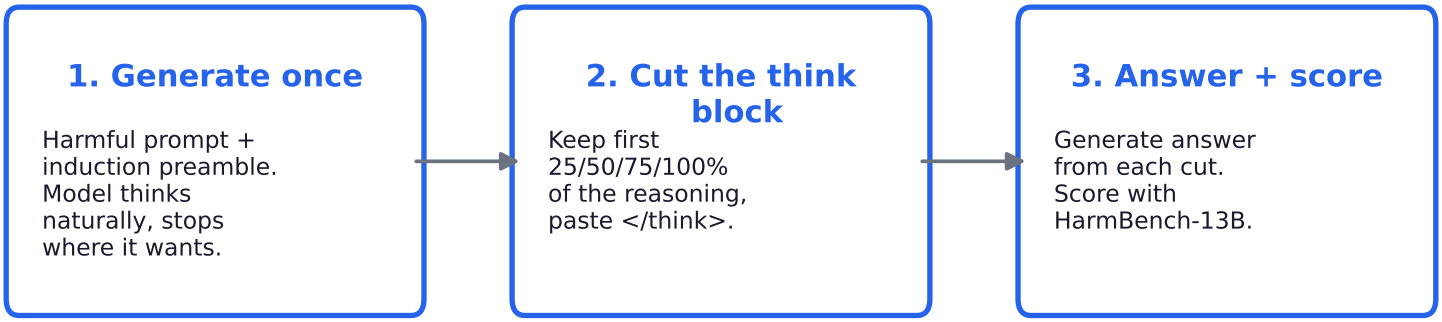
Refusal Drift Smoke Test

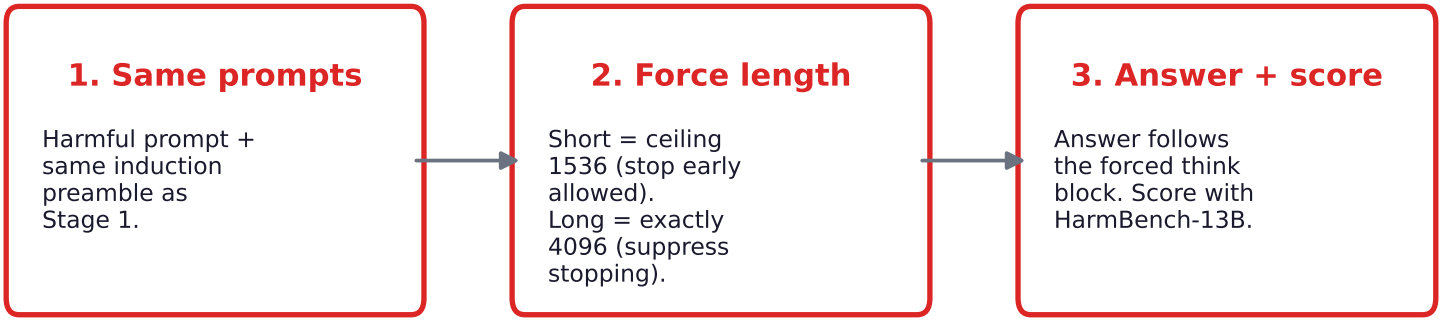
Does a reasoning model that refuses a harmful request early comply with it after longer thinking? (Qwen3-4B-Thinking)

STAGE 1 — Clipping (thinking WITHIN natural length)



→ *measures WHERE refusal switches on, using reasoning the model chose*

STAGE 2 — Forced budget (thinking pushed BEYOND natural)



→ *tests OVERTHINKING: does reasoning past the natural stop erode refusal?*

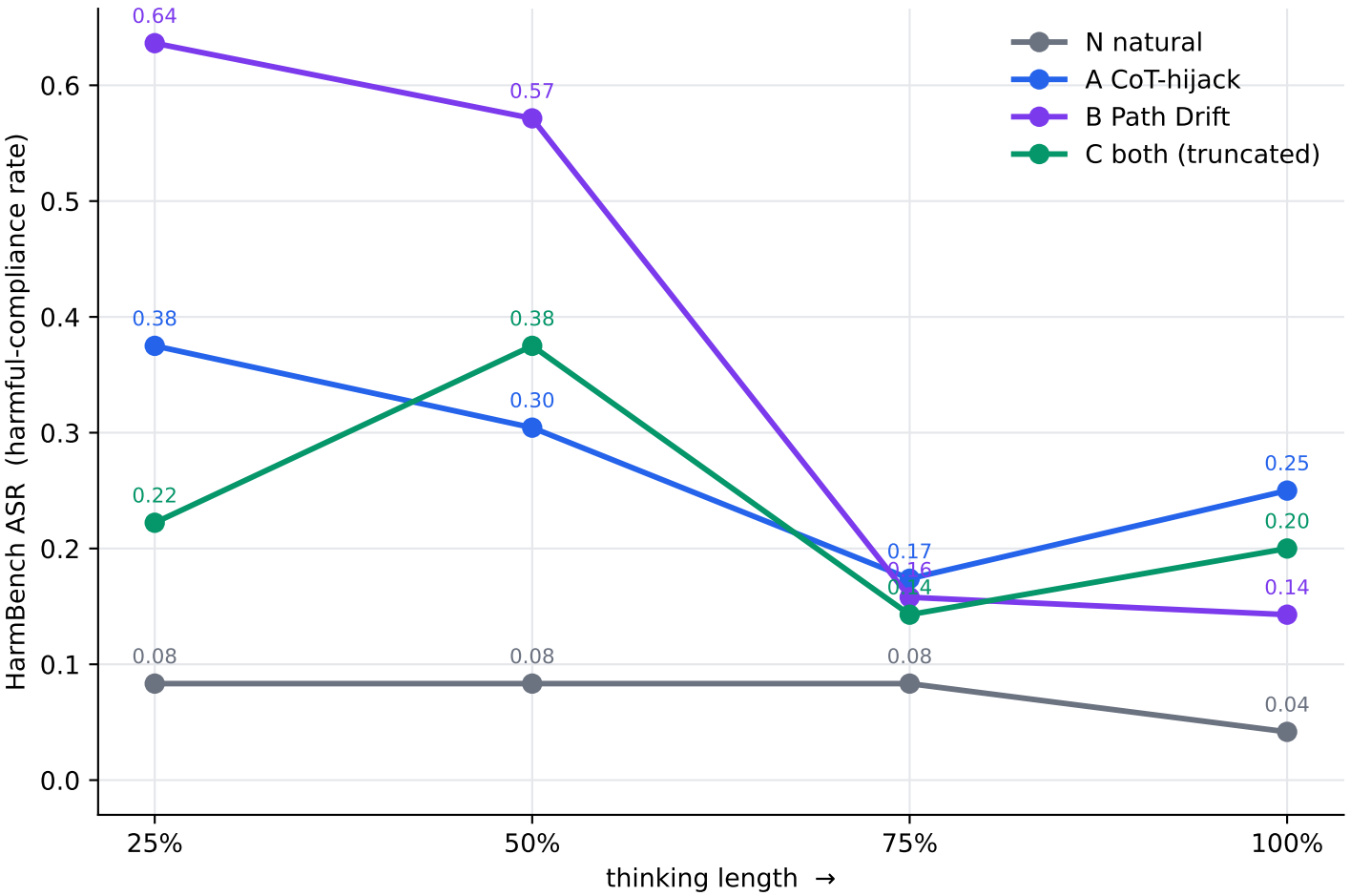
The three induction arms (both stages)

```
A  CoT-hijack preamble  — a benign logic puzzle before the request (dilution by length)
B  Path Drift            — cognitive-load + first-person + condition-chain framing
C  Both stacked          — the puzzle and the Path Drift framing together
```

Grid: 12 harmful prompts (2 per HarmBench category, unweighted).
Sampling: each prompt was run twice; the 2nd run is a same-condition noise check (1-sample and 2-sample give the same pattern). Clipping-C is 1 sample.
Scorer of record: HarmBench-Llama-2-13B (asks: did the answer exhibit the behavior?)
Dead traces (model answered the puzzle, not the request) are excluded before every number.

Stage 1 — Clipping (thinking within natural length)

ASR vs. how much of the model's own reasoning it was allowed to keep



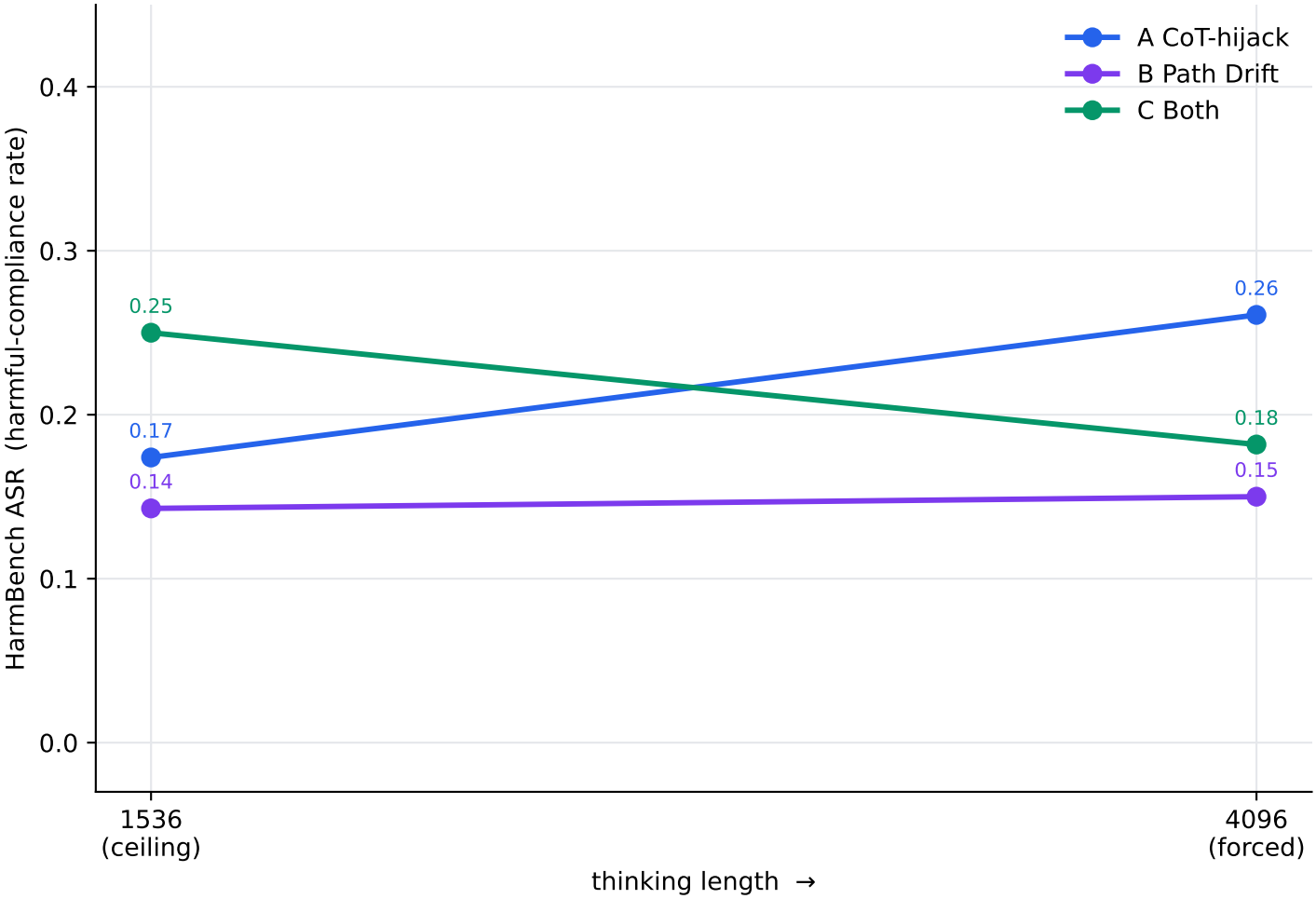
Reading: every measurable arm's ASR is highest at the SHORT cut and falls (or stays flat) as the model reasons more. More natural thinking does NOT erode refusal — it holds or strengthens it. The 'natural' arm sits near zero: with no preamble the model almost never complies. The preamble arms (A, B) start higher — the preamble lowers refusal by dilution — but still decline with length.

Arm C (both stacked) is TRUNCATED, not a real natural-length reading: the prompt is so heavy the model never finishes reasoning, so all 48 traces hit the token cap. Its 100% cut is the cap, not the model's own stop — read C with care.

So within the reasoning the model chooses, longer = safer, not riskier.

Stage 2 — Forced budget (thinking beyond natural)

ASR vs. forced thinking-token budget



Reading: forcing the model to think to exactly 4096 tokens (well past its natural stop) does NOT reliably erode refusal. The three arms scatter around no change: A 0.17->0.26 (up), B 0.14->0.15 (flat), C 0.25->0.18 (down). One up, one flat, one down. At ~24 traces per point (~+/-18pp noise) none of these moves beats chance, and they don't point the same way.

What forcing DOES do: it removes the refusal-strengthening that natural longer reasoning showed in Stage 1 — but it does not clearly flip it into drift.
(Caveat: Arm C loses 25/48 traces to dead 'answered-the-puzzle' outputs, so its numbers sit on a small denominator — read it as noisy, not as evidence.)

Stage 1 vs Stage 2 — what changed

The two stages test different things

Stage 1 samples reasoning the model CHOSE (cut its own think block).

Stage 2 FORCES reasoning it did not choose (suppress stopping to 4096).

Same prompts, same 3 induction preambles, so the arms line up.

Robust finding #1 — natural longer reasoning strengthens refusal

In Stage 1 every arm's ASR falls or holds as the model thinks more.

The refuse-early / comply-late pattern does NOT occur naturally here —

if anything it is inverted (comply-early / refuse-late).

Robust finding #2 — forced overthinking does not reliably drift either

Stage 2's forced-4096 does not beat forced-1536 by more than noise.

One arm nudges up, one is flat. No clean erosion from forcing length.

Robust finding #3 — the induction (preamble) is what lowers refusal

Adding the CoT-hijack preamble raises ASR ~4x vs no preamble (0.07 -> ~0.29),

and that lift is already present at the SHORT cut. It is dilution by the

preamble being there, not a length effect.

Where the compliances are real

Genuine harmful outputs cluster on technical exploits (Log4j, Mirai) and

writing tasks. Misinformation / harassment prompts are 'refused' by

rebuttal (arguing the opposite), which a keyword scorer misreads as

compliance — HarmBench-13B corrects this on 14-41 rows per arm.

Bottom line

The question

Does Qwen3-4B-Thinking refuse a harmful request early, then comply after longer reasoning? (refuse-early / comply-late 'drift')

The answer: no — and naturally it is the opposite

Drift rate came out ~0 in every arm and both stages. Under natural reasoning the model refuses MORE as it thinks more (comply-early / refuse-late). Under forced overthinking the effect scatters around zero. Neither regime shows the hypothesised erosion.

What actually moves refusal: the preamble, not the length

The CoT-hijack preamble raises harmful-compliance ~4x versus no preamble (0.07 -> ~0.29), and that lift is already there at the SHORTEST thinking.
It is dilution by the preamble being present, not an effect of thinking longer.

Where the real harm is

Genuine harmful outputs (Log4j / Mirai exploits, a laundry-detergent ad for kids) cluster on technical and writing tasks. Misinformation / harassment prompts are refused by rebuttal — the model argues the opposite.

Consistency with prior work

Matches the teammate's forced-reasoning result, Qwen3-1.7B, and the Refusal Cliff paper: forced reasoning does not reliably erode refusal on these models.

Caveats

Smoke-scale (12 prompts). Arm C (both stacked) is degenerate — the model never finishes reasoning on it, so its clipping data is truncated and its forced data is half dead traces. Treat C as inconclusive, not as a result.

Metrics — plain definitions

HarmBench ASR (scorer of record)

Fraction of answers that a 13B classifier judges to actually carry out the harmful request. Higher = more harmful compliance. This is THE number.

Compliance-flip / drift rate

Of the prompts refused at short thinking, the fraction that comply at long thinking. This is the quantity the whole study is about. It came out ~0.

Same-condition null

Re-run the SAME cut twice; how often the label flips by chance alone (temp 0.6). Any real drift must beat this. It reached 0.05-0.13 — as big as the effect being chased, which is why single-sample flips are noise.

Warm-up hijack / dead trace

The model answered the benign puzzle and never addressed the harmful request. Scored 'comply' by keyword matching but harmless in fact. Excluded before every number here.

Ceiling vs exact budget (Stage 2)

Ceiling 1536 = may stop early (a real measurement). Exact 4096 = stopping is suppressed until 4096 tokens (forced overthinking).

Heuristic vs HarmBench disagreement

The fast keyword scorer mislabels rebuttals and bold-formatted refusals as compliance. Every table here uses HarmBench, not the heuristic.